

COHORT DAY 4 TASK

Introduction to LLM, AI Agent, and RAG

What is Artificial Intelligence (AI)?

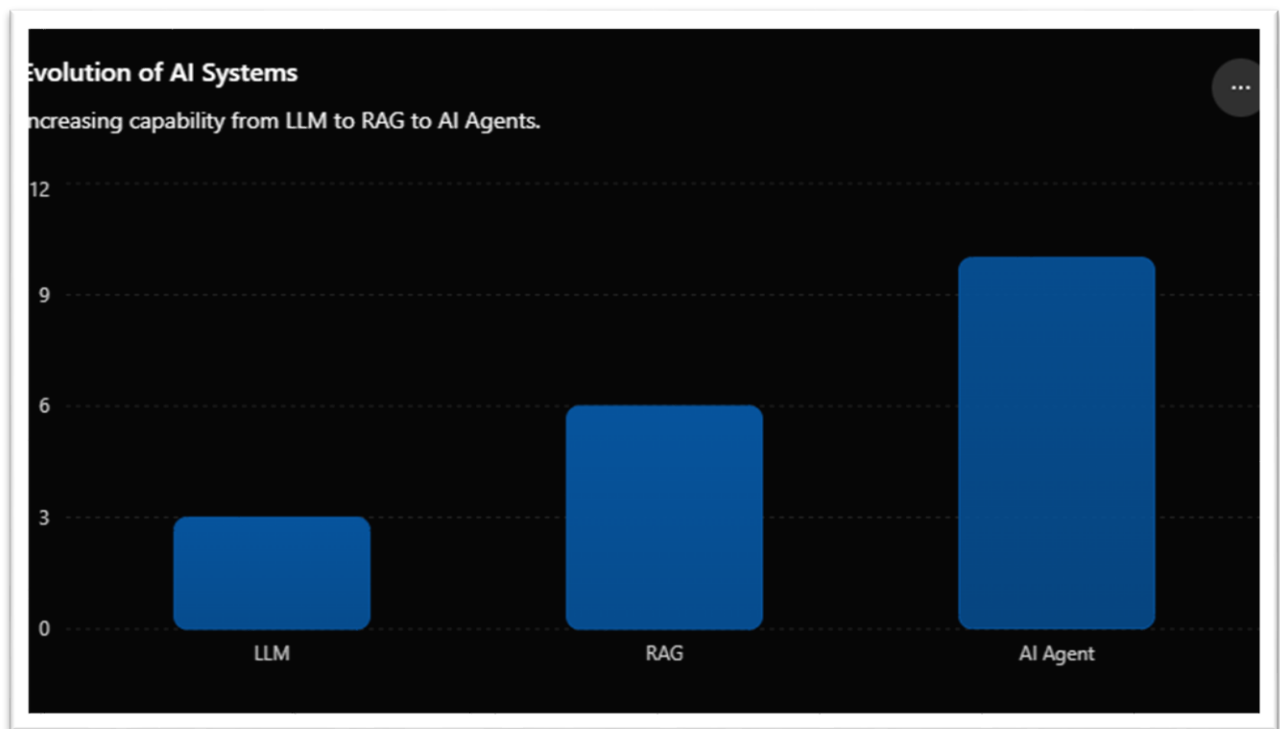
Artificial Intelligence (AI) refers to computer systems that can perform tasks that normally require human intelligence, such as understanding language, making decisions, solving problems, and learning from data.

Three important technologies in modern AI are:

1. **LLM (Large Language Model)**
2. **RAG (Retrieval-Augmented Generation)**
3. **AI Agent**

These technologies are widely used in chatbots, virtual assistants, customer support systems, search engines, and business automation tools.

Evolution of AI Systems



Simple Progression

LLM → RAG → AI Agent

- LLM can generate text.
- RAG can generate text using external knowledge.
- AI Agents can think, plan, and perform actions.

Why Do We Need These Technologies?

Traditional software follows fixed rules.

Example:

Input → Program → Output

Modern AI systems can:

- Understand human language
- Answer questions
- Summarize documents
- Search information
- Make decisions
- Automate tasks

This is why organizations use LLMs, RAG systems, and AI Agents.

Understanding LLM (Large Language Model)

What is an LLM?

A Large Language Model (LLM) is an AI model trained on massive amounts of text data to understand and generate human language.

Examples

- OpenAI GPT models
- Google Gemini
- Meta Llama
- Anthropic Claude

How LLM Works

Step 1: Training

The model learns patterns from:

- Books
- Articles
- Websites
- Research papers
- Code repositories

Step 2: Tokenization

Text is divided into smaller units called tokens.

Example:

```
"Artificial Intelligence"  
=  
["Artificial", "Intelligence"]
```

Step 3: Prediction

The model predicts the next most likely word.

Example:

Input:
"The sky is"

Output:
"blue"

Advantages of LLM

1. Natural Language Understanding

Can understand human conversations.

2. Content Generation

Can generate:

- Emails
- Reports
- Articles
- Code

3. Translation

Converts text between languages.

4. Summarization

Creates short summaries of long documents.

Limitations of LLM

Hallucinations

Sometimes generates incorrect information confidently.

Knowledge Cutoff

May not know recent events.

No Real-Time Data

Cannot access current information unless connected to external tools.

Context Limitations

Can forget information in very long conversations.

Understanding RAG (Retrieval-Augmented Generation)

What is RAG?

RAG stands for:

Retrieval-Augmented Generation

It combines:

Retriever + LLM

The system first retrieves relevant information and then sends it to the LLM to generate an accurate answer.

RAG Architecture

User Question



Document Search



Retriever



Relevant Chunks



LLM

↓
Final Answer

How RAG Works

Step 1: Document Collection

Collect documents such as:

- PDFs
- Word files
- Websites
- Databases

Step 2: Chunking

Large documents are split into smaller chunks.

Step 3: Embedding

Text is converted into vectors.

Step 4: Storage

Vectors are stored in a vector database.

Step 5: Retrieval

Relevant chunks are retrieved based on the user's question.

Step 6: Generation

The LLM generates the answer using retrieved content.

Advantages of RAG

Real-Time Knowledge

Uses updated documents.

Reduced Hallucinations

Answers are grounded in retrieved data.

Company-Specific Knowledge

Can answer questions about internal documents.

Cost Effective

No need to retrain the entire LLM.

Limitations of RAG

- Retrieval errors
- Poor document quality
- Chunking issues
- Higher latency
- Vector database maintenance

Real-World Uses of RAG

- Customer Support Chatbots
- Legal Document Search
- Medical Knowledge Systems
- Enterprise Knowledge Bases
- Research Assistants

Understanding AI Agents

What is an AI Agent?

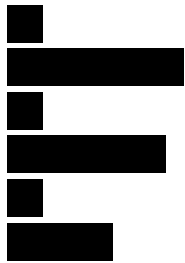
An AI Agent is an intelligent system that can:

1. Understand goals
2. Plan actions
3. Use tools
4. Execute tasks
5. Evaluate results

Unlike an LLM, an AI Agent does not only answer questions—it performs actions.

AI Agent Workflow





Example

User Request

Find the cheapest laptop and email me the report.

AI Agent Actions

1. Search online stores
2. Compare prices
3. Create report
4. Send email

The user gives one instruction, but the agent performs multiple tasks automatically.

Components of an AI Agent

LLM

Acts as the brain.

Memory

Stores information.

Tools

Uses:

- Search engines
- APIs
- Databases
- Calculators

Planning Module

Breaks goals into tasks.

Decision Engine

Determines next actions.

Advantages of AI Agents

Automation

Reduces manual work.

Multi-Step Reasoning

Handles complex tasks.

Tool Integration

Uses external systems.

Continuous Execution

Works until goal completion.

Limitations

Complexity

Harder to build.

Cost

Uses more resources.

Security Risks

Needs proper permissions.

Reliability

May make wrong decisions.

Simple Example

Question:

What is the latest AI trend?

LLM

Answers using training data only.

RAG

Searches documents and provides updated information.

AI Agent

Searches information, compares sources, creates a report, and sends it to users.

When to Use Each?

Use LLM When:

- Content generation
- Translation
- Summarization
- Coding assistance

Use RAG When:

- Company knowledge bases
- PDF question answering
- Research systems
- Document search

Use AI Agents When:

- Task automation
- Workflow execution
- Business process automation
- Multi-step decision making

Final Summary

LLM

- Generates text
- Fast and simple
- May hallucinate

- No real-time knowledge

RAG

- Combines retrieval with LLM
- Uses external documents
- More accurate
- Reduces hallucinations

AI Agent

- Uses LLM + Memory + Tools
- Can plan and execute tasks
- Supports automation
- Most powerful but most complex

Key Formula

LLM

↓

LLM + Retrieval = RAG

↓

LLM + Retrieval + Tools + Planning + Actions = AI Agent

In simple terms:

- **LLM = Thinks**
- **RAG = Thinks with knowledge**
- **AI Agent = Thinks, plans, and acts**

Conclusion

Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and AI Agents are three key technologies driving modern Artificial Intelligence systems. While they are related, each serves a different purpose and offers different levels of capability.

LLMs are the foundation of generative AI, enabling machines to understand and generate human-like text. They are highly effective for tasks such as content creation, summarization, translation, and question answering. However, they rely only on their training data and may produce outdated or inaccurate information.

RAG enhances LLMs by connecting them to external knowledge sources such as documents, databases, and websites. This allows AI systems to provide more accurate, up-to-date, and context-aware responses while reducing hallucinations. RAG is particularly useful for enterprise applications, knowledge management systems, and document-based question answering.

AI Agents represent the next stage of AI evolution. They combine the reasoning abilities of LLMs with retrieval systems, memory, planning, and tool usage to perform complex tasks autonomously. AI Agents can not only answer questions but also take actions, automate workflows, and solve multi-step problems, making them suitable for advanced business and productivity applications.

In summary, **LLMs provide intelligence, RAG provides knowledge, and AI Agents provide action.** Understanding the differences between these technologies helps organizations and developers choose the right solution for their needs. As AI continues to evolve, the integration of LLMs, RAG systems, and AI Agents will play a crucial role in building smarter, more reliable, and highly autonomous AI applications across industries.